LAB – 4

Project Title: Adaptive-Software-Development-Lab
Project Scenario: Online Course Registration System (OCRS)

PHASE 1: UNDERSTANDING ADAPTIVE SOFTWARE DEVELOPMENT

Step 1: Study ASD Characteristics

1. Mission-Focused Development

Mission-focused development means the team concentrates on achieving the main goal of the project instead of strictly following a fixed plan. The focus is on delivering value to users. This allows the team to adjust tasks and priorities if requirements change.

2. Component-Based Development

In ASD, software is developed in small components or modules. Each component performs a specific function and can be developed independently. This approach makes the system flexible and allows easy modification or improvement.

3. Time-Boxed Iterations

ASD uses fixed time periods called iterations to complete tasks. Each iteration has a limited duration, which encourages teams to deliver results quickly. Time-boxing helps maintain progress even when requirements change.

4. Risk-Driven and Change-Tolerant Planning

ASD accepts that changes and risks are part of software development. Instead of avoiding them, teams plan in a way that allows easy adaptation. This helps the project respond effectively to new information and unexpected challenges.

5. Continuous Learning

Continuous learning is an important part of ASD. After each iteration, the team reviews the results and learns from successes and mistakes. This learning helps improve future planning and development decisions.

Why Traditional Planning Fails in Uncertain Environments

Traditional planning assumes that all requirements are known at the beginning of the project. However, in real software projects, requirements often change due to new technology, user needs, or market conditions. When plans are fixed and rigid, teams cannot easily adapt to these changes. As a result, traditional planning may lead to delays, increased costs, or products that do not meet user expectations.

Step 2: Understand the SCL Cycle

1. Speculate

In a real-world project like an online shopping application, the development team may assume that users prefer quick checkout and multiple payment options. Based on this assumption, they plan features such as guest checkout and digital payments.

2. Collaborate

During development, developers, designers, testers, and stakeholders work together to discuss features and solve problems. For example, the development team collaborates with business managers to decide how the product search feature should work.

3. Learn

After releasing a feature, the team collects feedback from users. For example, if users report difficulty in using the payment page, the team learns from this feedback and improves the design in the next iteration.

PHASE 2: SPECULATE PHASE USING TAIGA

Step 3: Create ASD Project in Taiga